

VAR

$$Y_t = C + A_1 Y_{t-1} + A_2 Y_{t-2} + \dots + A_p Y_{t-p} + \varepsilon_t$$

where $Y_t \in \mathbb{R}^{k \times 1}$, $A_i \in \mathbb{R}^{k \times k}$

$$\cancel{VAR} (I - A_1 L - A_2 L^2 - \dots - A_p L^p) Y_t = C + \varepsilon_t$$

Characteristic equation:-

$$\det(I - A_1 z - A_2 z^2 - \dots - A_p z^p) = 0$$

Stationary $\Rightarrow$ Roots lies outside the unit circle
i.e. $|z| > 1$

~~Not~~ But, computationally expensive,

$$\text{Let } Z_t = \begin{bmatrix} Y_t \\ Y_{t-1} \\ \vdots \\ Y_{t-p+1} \end{bmatrix} \in \mathbb{R}^{kp \times 1}$$

$$Z_t = F Z_{t-1} + V_t, \text{ where } F = \begin{bmatrix} A_1 & A_2 & \dots & A_p \\ I & 0 & \dots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \dots & I & 0 \end{bmatrix}$$

i.e. $\text{var}(p)$ converted to $\text{var}(1)$

~~now for stationary~~

here, stationary $\Leftrightarrow$ eigen values of F matrix are less than 1.

~~Prop 1~~:- if one/more eigen values are greater ^{/equal} than 1.

Mathematical symptom of a unit root in ~~the system~~ the system

checking -

~~to~~ To confirm it,

① Apply ADF test on each variable

~~(y_{1t}, y_{2t}, ..., y_{kt})~~

H_0 : series has unit root (i.e. non-stationary)

$\rightarrow (y_{11}, y_{12}, \dots, y_{1k})$

t=0	y ₁₀	y ₂₀	...	y _{k0}
t=1				
t=T				

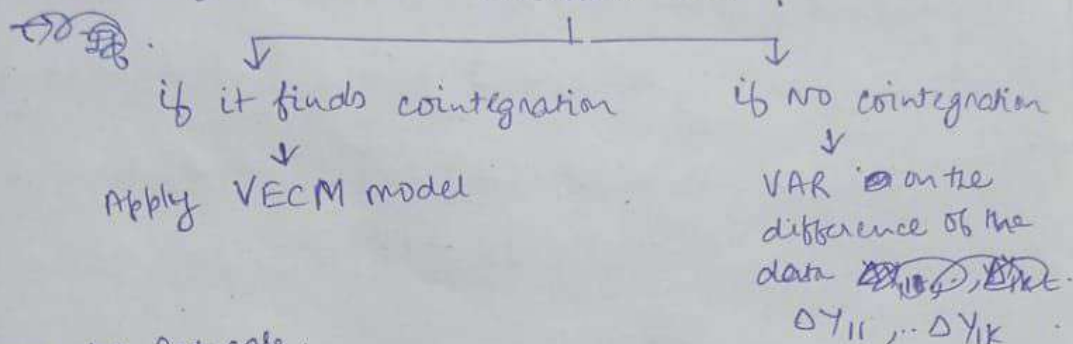
Applying ADF test on each column, ~~and~~ there are 3 cases :-

① All variables are $I(0)$:

→ proceed to VAR, estimate & forecast.

② All variables are $I(1)$:

→ cointegration test: Johansen test.



③ Ignore the 3rd case.

Suppose, data $\rightarrow$ non-stationary ($I(1)$).

If we regress one $I(1)$ variable with another result $\Rightarrow$ spurious (we can get high R^2 score & significant t -values) ~~too~~ even if they have no meaningful relationship.

When all variables are $I(1) \rightarrow$ but a non-trivial linear combination $\beta' Z_t$ is $I(0)$.

$\beta \rightarrow$ cointegrating vector (specific, stationary relationship b/w the variables).

If system has K variables & m cointegrating vectors then $K-m$ common (stochastic) trends driving the system.

If we just difference all ~~data~~ data to make it stationary & build VAR on that $\rightarrow$ we throw out the information of long run equilibrium.
Over differencing can make model invertebrate.

2. var(z) model.

Let $p=2$

$$z_t = c_0 + \phi_1 z_{t-1} + \phi_2 z_{t-2} + a_t$$

To make it stationary, we difference it 1 time,

$$\Delta z_t = \phi_1 \Delta z_{t-1}$$

$$z_t - z_{t-1} = c_0 + \phi_1 z_{t-1} - z_{t-1} + \phi_2 z_{t-2} + a_t$$

$$\Rightarrow \Delta z_t = c_0 + (\phi_1 - I) z_{t-1} + \phi_2 z_{t-2} + a_t$$

$$\Rightarrow \Delta z_t = c_0 + (\phi_1 - I) z_{t-1} + \phi_2 z_{t-2} - \phi_2 z_{t-1} + \phi_2 z_{t-1} + a_t$$

$$= c_0 + \cancel{(\phi_1 - I) z_{t-1}} - \phi_2 (z_{t-1} - z_{t-2}) + (\phi_1 + \phi_2 - I) z_{t-1} + a_t$$

$$\therefore \Delta z_t = \cancel{c_0} + \pi z_{t-1} + \bar{\Phi}_1^* \Delta z_{t-1} + a_t$$

$$\text{where } \pi = (\phi_1 + \phi_2 - I), \bar{\Phi}_1^* = -\phi_2$$

In General,

VECM

$$\Delta z_t = c(t) + \pi z_{t-1} + \sum_{i=1}^{p-1} \bar{\Phi}_i^* \Delta z_{t-i} + a_t$$

$$\text{where } \pi = (\phi_1 + \phi_2 - I)$$

$$\pi = -(I - \phi_1 - \phi_2 - \dots - \phi_p)$$

$$\bar{\Phi}_i^* = -(\phi_{i+1} + \phi_{i+2} + \dots + \phi_p)$$

~~$\Delta z_t \rightarrow \text{stationary}$~~ $\Delta z_t, \Delta z_{t-1}, a_t \rightarrow \text{stationary}$
 ~~$\Delta z_t \rightarrow \text{stationary}$~~ $c(t) \rightarrow \text{deterministic trend}$ ^{constant/}

$$\Pi z_{t-1} \rightarrow I(1) \longrightarrow$$

we have to make this stationary, it becomes stationary only if " Π " is cointegrating matrix with a reduced rank m .

A matrix with reduced rank can be split into two smaller matrices α & β'

so, replacing Π by $\alpha\beta'$

$$\text{or get, } (\alpha\beta') z_{t-1} = \alpha(\beta' z_{t-1})$$

From defⁿ of cointegration,

$\beta' z_{t-1}$ is stationary.

Thus it becomes stationary.

now, we cannot use OLS, to estimate the parameters (as it is non-linear $(\alpha\beta')$)

To get all the coefficients is Johansen procedure i.e. MLE approach.

we have to find β' first,

$$\Pi z_{t-1} \rightarrow \text{long run parts} \left| \sum_{i=1}^{p-1} \Phi_i^* \Delta z_{t-1} \rightarrow \text{short run parts.} \right.$$

To find long run β we have to eliminate short run.

we regress Δz_t on all the short run lagged $\Delta z_{t-1}, \Delta z_{t-2}, \dots$

We save the residuals, let call it R_0 .

($R_0 \rightarrow$ part of ΔZ_t that ~~can't~~ can't be explained by short run dynamics i.e. it is the pure long ~~run~~ ^{run} movement)

$$R_0 : \{R_{01}, R_{02}, \dots, R_{0T}\}$$

residual $k \times 1$ vector

We regress Z_{t-1} on all $\Delta Z_{t-1}, \Delta Z_{t-2}, \dots$

Residuals = R_1 (R_1 is the part of Z_{t-1} not related to short run).

$$R_1 : \{R_{11}, R_{12}, \dots, R_{1T}\}$$

So, VECM boils down to,

$$R_0 = (\alpha\beta')R_1 + \text{error}$$

Johansen test

→ ① are ~~the~~ ^{the} $I(1)$ variables ~~can~~ cointegrated?

②. If so, ~~how~~ how many cointegrating relationships are there?

Test - for the rank of Π matrix.

Rank = 0 $\rightarrow$ ~~no~~ Π zero matrix $\rightarrow$ no cointegration.

Rank = $m \rightarrow m$ cointegrating relationships.

VECM eqn. : $R_0 = \pi R_1 + \epsilon$ with ϵ .

covariance between R_0 & R_1 :

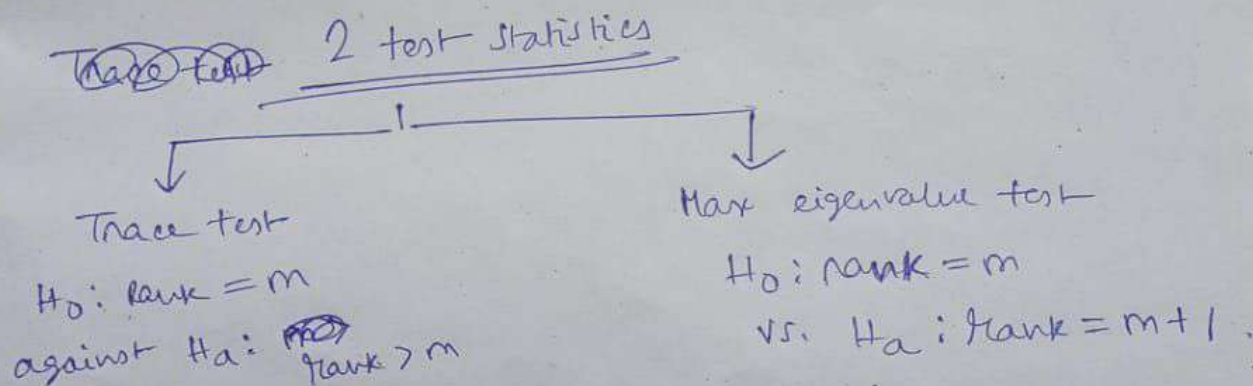
$K \times K$ matrix $\left\{ \begin{array}{l} \hat{\Sigma}_{00} \rightarrow \text{sample covariance matrix of } R_0 \\ \hat{\Sigma}_{11} \rightarrow \text{" " " of } R_1 \\ \hat{\Sigma}_{01} \rightarrow \text{" cross " " of } R_0 \& R_1 \end{array} \right.$

using CCA (canonical correlation analysis)

$$\Sigma_{11}^{-1} \Sigma_{10} \Sigma_{00}^{-1} \Sigma_{01} g_i = \lambda_i g_i$$

Each eigenvalue represents the strength of a possible cointegrating relationship.

→ Rank m of π is the number of non-zero eigenvalues.



Starting from $m=0$,

if H_0 is rejected, then test for $m=1$ and so on.

~~until~~ we stop at the first m , for which H_0 is not

rejected -

After getting m ,

$\hat{\beta}' = \begin{pmatrix} -g_1 \\ -g_2 \\ \vdots \\ -g_m \end{pmatrix}$ where $g_i \rightarrow$ corresponding eigenvectors of λ_i 's.

After getting $\hat{\beta}$, other parameters can be estimated by simple ~~and~~ OLS regression.